Lec 3. Sampling distributions of order statistics

Zhang W.P.

Contents

1.1 Order statistics 1
1.2 Distribution of sample range 9
1.3 Properties of sample quantile 11

Learning objectives

After this section, you should be able to ...

- derive the distribution of order statistics
- understand the properties of sample quantile

1.1 Order statistics

These statistics have in recent times come to play an important role in statistical inference partly because some of their properties do not depend upon the distribution from which the random sample is obtained.

Definition

Let $X_1, X_2, \ldots, X_n$ denote a random sample from a distribution of the continuous type having a pdf $f(x)$ that has support $\mathcal{S} = (a, b)$, where $-\infty \leq a < b \leq \infty$. Let $X_{(1)}$ be the smallest of these X_i, $X_{(2)}$ the next X_i in order of magnitude, $\ldots$, and $X_{(n)}$ the largest of X_i. That is, $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$ represent $X_1, X_2, \ldots, X_n$ when the latter are arranged in ascending order of magnitude. We call $X_{(i)}$, $i = 1, 2, \ldots, n$, the ith order statistic of the random sample $X_1, X_2, \ldots, X_n$.

Theorem 1. *Using the above notation, let* $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$ *denote the* n *order statistics based on the random sample* $X_1, X_2, \ldots, X_n$ *from a continuous distribution with pdf* $f(x)$ *and support* (a, b)*. Then the joint pdf of* $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$ *is given by*

$$g(y_1, y_2, \ldots, y_n) = \begin{cases} n! f(y_1) f(y_2) \cdots f(y_n) & a < y_1 < y_2 < \cdots < y_n < b \\ 0 & \textit{elsewhere.} \end{cases}$$

Example

Let X denote a random variable of the continuous type with a pdf $f(x)$ that is positive and continuous, with support $\mathcal{S} = (a, b), -\infty \leq a < b \leq \infty$. The distribution function $F(x)$ of X may be written

$$F(x) = \int_a^x f(w)dw, \quad a < x < b$$

If $x \leq a, F(x) = 0$; and if $b \leq x, F(x) = 1$. Thus there is a unique median m of the distribution with $F(m) = \frac{1}{2}$. Let X_1, X_2, X_3 denote a random sample from this distribution and let $Y_1 < Y_2 < Y_3$ denote the order statistics of the sample. Note that Y_2 is the sample median. We compute the probability that $Y_2 \leq m$.

Solution. The joint pdf of the three order statistics is

$$g(y_1, y_2, y_3) = \begin{cases} 6f(y_1)f(y_2)f(y_3) & a < y_1 < y_2 < y_3 < b \\ 0 & \text{elsewhere} \end{cases}$$

The pdf of Y_2 is then

$$\begin{aligned} h(y_2) &= 6f(y_2) \int_{y_2}^{b} \int_{a}^{y_2} f(y_1)f(y_3)dy_1 dy_3 \\ &= \begin{cases} 6f(y_2)F(y_2)\left[1 - F(y_2)\right] & a < y_2 < b \\ 0 & \text{elsewhere} \end{cases} \end{aligned}$$

Accordingly,

$$\begin{aligned} P(Y_2 \leq m) &= 6 \int_{a}^{m} \left\{F(y_2)f(y_2) - \left[F(y_2)\right]^2 f(y_2)\right\} dy_2 \\ &= 6 \left\{ \frac{\left[F(y_2)\right]^2}{2} - \frac{\left[F(y_2)\right]^3}{3} \right\}_a^m = \frac{1}{2} \end{aligned}$$

Hence, for this situation, the median of the sample median Y_2 is the population median m. ■

Marginal distribution

Once it is observed that

$$\int_a^x [F(w)]^{\alpha-1} f(w)dw = \frac{[F(x)]^\alpha}{\alpha}, \quad \alpha > 0$$

and that

$$\int_y^b [1-F(w)]^{\beta-1} f(w)dw = \frac{[1-F(y)]^\beta}{\beta}, \quad \beta > 0$$

it is easy to express the marginal pdf of any order statistic, say Y_k, in terms of $F(x)$ and $f(x)$. This is done by evaluating the integral

$$g_k(y_k) = \int_a^{y_k} \cdots \int_a^{y_2} \int_{y_k}^b \cdots \int_{y_n-1}^b n! f(y_1) f(y_2) \cdots f(y_n) dy_n \cdots dy_{k+1} dy_1 \cdots dy_{k-1}$$

The result is

$$g_k(y_k) = \begin{cases} \frac{n!}{(k-1)!(n-k)!} [F(y_k)]^{k-1} [1-F(y_k)]^{n-k} f(y_k) & a < y_k < b \\ 0 & \text{elsewhere.} \end{cases} \tag{1.1}$$

Finally, the joint pdf of any two order statistics, say $Y_i < Y_j$, is easily expressed in terms of $F(x)$ and $f(x)$. We have

$$g_{ij}(y_i, y_j) = \int_a^{y_i} \cdots \int_a^{y_2} \int_{y_i}^{y_j} \cdots \int_{y_{j-2}}^{y_j} \int_{y_j}^{b} \cdots \int_{y_{n-1}}^{b} n! f(y_1) \times \cdots$$
$$\times f(y_n) dy_n \cdots dy_{j+1} dy_{j-1} \cdots dy_{i+1} dy_1 \cdots dy_{i-1}.$$

Since, for $\gamma > 0$

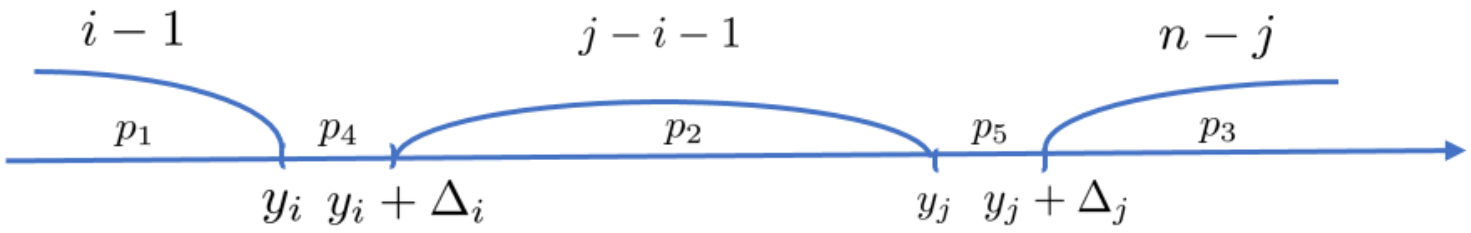
$$\int_x^y [F(y) - F(w)]^{\gamma-1} f(w) dw = - \left. \frac{[F(y) - F(w)]^\gamma}{\gamma} \right|_x^y$$
$$= \frac{[F(y) - F(x)]^\gamma}{\gamma},$$

it is found that

$$g_{ij}(y_i, y_j) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} [F(y_i)]^{i-1} [F(y_j) - F(y_i)]^{j-i-1}$$
$$\times [1 - F(y_j)]^{n-j} f(y_i) f(y_j) I(a < y_i < y_j < b) \tag{1.2}$$

Heuristic Derivation

There is an easy method of remembering the pdf of a vector of order statistics such as the one given in formula (1.2). The probability $P(y_i < Y_i < y_i + \Delta_i, y_j < Y_j < y_j + \Delta_j)$, where Δ_i and Δ_j are small, can be approximated by the following multinomial probability. In n independent trials,



where $p_1 = F(y_i)$, $p_2 = F(y_j) - F(y_i)$, $p_3 = 1 - F(y_j)$, $p_4 = f(y_i)\Delta_i$ and $p_5 = f(y_j)\Delta_j$. This multinomial probability is

$$\frac{n!}{(i-1)!(j-i-1)!(n-j)!1!1!} p_1^{i-1} p_2^{j-i-1} p_3^{n-j} p_4 p_5$$

which is $g_{i,j}(y_i, y_j)\Delta_i\Delta_j$, where $g_{i,j}(y_i, y_j)$ is given in expression (1.2)

Example

Let $X_1, \ldots, X_n$ be a sample from $U(0,1)$, and $X_{(1)}, \ldots, X_{(n)}$ be its order statistic, then

(1) $X_{(i)} \sim Beta(i, n-i+1)$ with density

$$f_i(x) = \frac{n!}{(i-1)!(n-i)!} x^{i-1}(1-x)^{n-i} I(0 < x < 1).$$

(2) for $i < j$, the joint density of $(X_{(i)}, X_{(j)})$ is

$$f_{ij}(x,y) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} x^{i-1}(y-x)^{j-i-1}(1-y)^{n-j} I(0 < x < y <$$

1.2 Distribution of sample range

For sample range(样本极差) $\mathrm{R}_n = X_{(n)} - X_{(1)}$, because the joint density of $X_{(1)}$ and $X_{(n)}$ is given by

$$f_{X_{(1)},X_{(n)}}(x,y) = n(n-1)(F(y)-F(x))^{n-2} f(x)f(y)I(x<y),$$

the density of R_n can be shown as

$$f_{\mathrm{R}_n}(r) = n(n-1)\int (F(z+r)-F(z))^{n-2} f(z)f(z+r)dz \tag{1.3}$$

Example

If $X \sim U(0,1)$, the density (1.3) becomes

$$f_{\mathrm{R}_n}(r) = n(n-1)\int_0^{1-r}(z+r-z)^{n-2}dz = n(n-1)r^{n-2}(1-r)$$

for $0 < r < 1$ and $f_{\mathrm{R}_n}(r) = 0$ otherwise.

1.3 Properties of sample quantile

Recall

Definition

For $0 < p < 1$, the pth quantile (分位数) of F is defined as

$$\xi(p) = F^{-1}(p) = \inf\{x : F(x) \geq p\}$$

- If F is continuous and strictly increasing, F^{-1} is defined by

$$F^{-1}(y) = x \quad \text{when } y = F(x)$$

- If F has a discontinuity at x_0, suppose that $F(x_0-) < y < F(x_0)$. In this case, although there exists no x for which $y = F(x), F^{-1}(y)$ is defined to be equal to x_0
- If $U \sim U(0,1)$, then $X = F_X^{-1}(U) \sim F_X(x)$.
- If $F(x) = G(\frac{x-\mu}{\sigma})$, G is strictly increasing, then $F^{-1}(p) = \mu + \sigma G^{-1}(p)$.

Definition

Let $\{X_1, X_2, \ldots, X_n\}$ be a sample from F, the sample pth quantile $\hat{\xi}_{pn}$ (样本 p 分位数) (which is $X_{([np])}$ or $X_{([np]+1)}$ depending on whether np is an integer, here $[np]$ denotes the integer part of np) is defined as

$$\hat{\xi}_{pn} = F_n^{-1}(p) = \inf\{x : F_n(x) \geq p\}$$

where F_n is the corresponding empirical distribution function.

- $X_{(i)} = F_n^{-1}\left(\frac{i}{n}\right), i = 1, \ldots, n$
- $\{\hat{\xi}_{pn} > c\} \Leftrightarrow \{F_n(c) < p\}$
- $\{\hat{\xi}_{pn} < c\} \Leftrightarrow \{p \leq F_n(c)\}$

Theorem 2. *Let $0 < p < 1$. Suppose that ξ_p is the unique solution x of $F(x-) \leq p \leq F(x)$. Then, the pth sample quantile $\hat{\xi}_{pn} \xrightarrow{P} \xi_p$.*

Proof. Let $\epsilon > 0$. Write

$$P\left(\left|\hat{\xi}_{pn} - \xi_p\right| > \epsilon\right) = P\left(\hat{\xi}_{pn} > \xi_p + \epsilon\right) + P\left(\hat{\xi}_{pn} < \xi_p - \epsilon\right)$$

Since

$$\begin{aligned} P\left(\hat{\xi}_{pn} > \xi_p + \epsilon\right) &= P\left(p > F_n\left(\xi_p + \epsilon\right)\right) \\ &= P\left(\sum_{i=1}^{n} I\left(X_i > \xi_p + \epsilon\right) > n(1-p)\right) \\ &= P\left(\sum_{i=1}^{n} V_i - \sum_{i=1}^{n} E\left(V_i\right) > n\delta_1\right) \end{aligned}$$

where $V_i = I\left(X_i > \xi_p + \epsilon\right)$ and $\delta_1 = F\left(\xi_p + \epsilon\right) - p$. Likewise,

$$P\left(\hat{\xi}_{pn} < \xi_p - \epsilon\right) = P\left(p \leq F_n\left(\xi_p - \epsilon\right)\right) = P\left(\sum_{i=1}^{n} W_i - \sum_{i=1}^{n} E\left(W_i\right) \geq n\delta_2\right)$$

where $W_i = I\left(X_i \leq \xi_p - \epsilon\right)$ and $\delta_2 = p - F\left(\xi_p - \epsilon\right)$. Therefore, utilizing Chebyshev's inequality we can prove it. $\square$

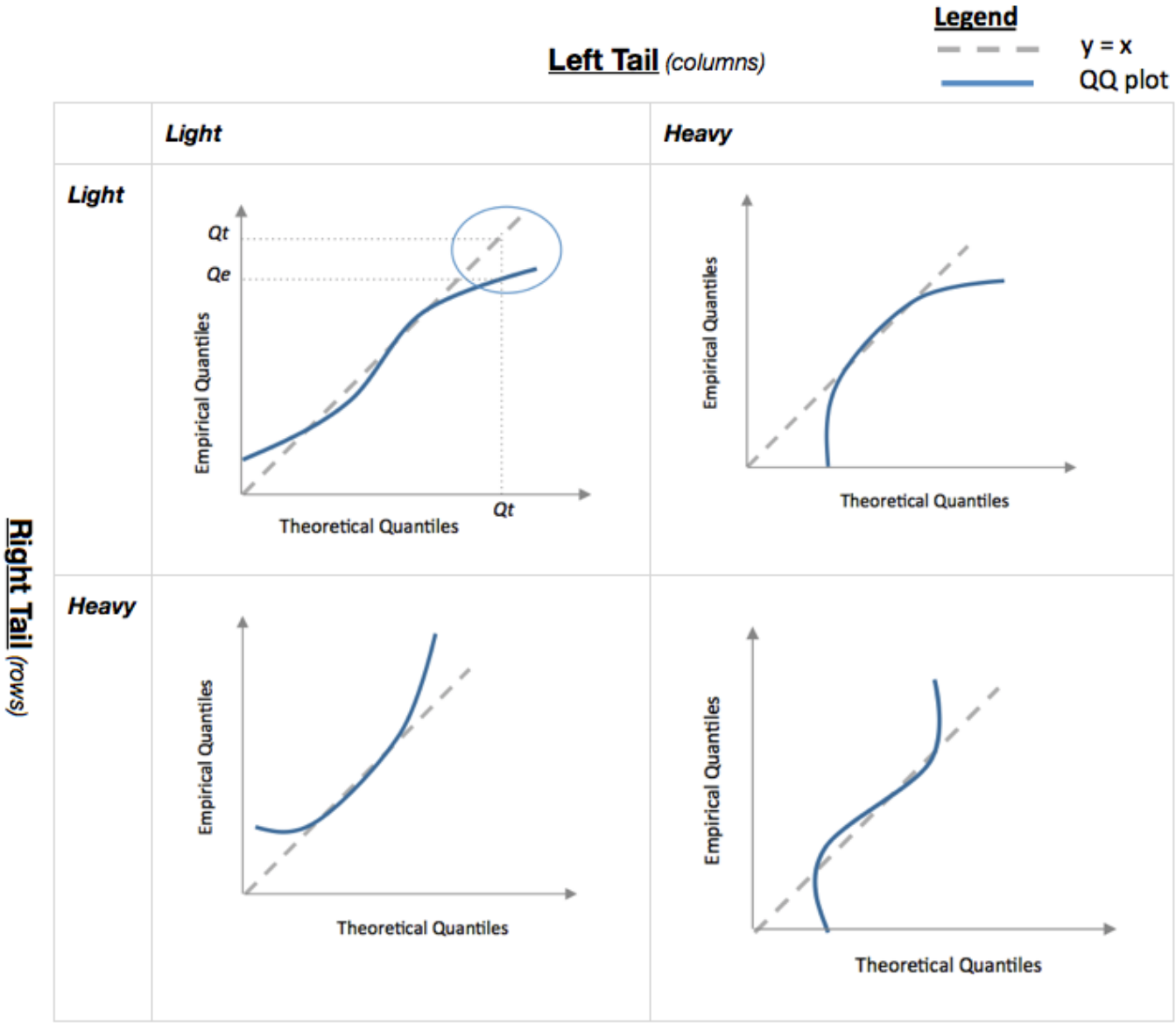
Probability Plots

A *probability plot*, also called a quantile-quantile plot or a **Q-Q plot** for short, is an informal, graphical test to determine if observed data come from a specified distribution.

Let $x_1, \ldots, x_n$ $i.i.d \sim F$, and the order statistics be $\left(x_{(1)}, x_{(2)}, \ldots, x_{(n)}\right)$. If the data really do come from the distribution, then we would expect the points

$$\left(x_{(i)}, F^{-1}\left(\frac{i}{n+1}\right)\right), i = 1, \ldots, n$$

to be close to the diagonal line $y = x$; conversely, strong deviation from this line is evidence that the distribution did not produce the data. The plot of these points is referred to as a probability plot.


Legend
y = x
QQ plot
Left Tail (columns)
Right Tail (rows)
Light
Heavy
Light
Heavy
Qt
Qe
Qt
Empirical Quantiles
Theoretical Quantiles

Theorem 3. *Let $0 < p < 1$. Suppose that F possesses a density f in a neighborhood of ξ_p and f is positive and continuous at ξ_p, then the pth sample quantile $\hat{\xi}_{pn}$ satisfies that*

$$\sqrt{n}\left(\hat{\xi}_{pn} - \xi_p\right) \xrightarrow{d} N\left(0, \frac{p(1-p)}{\left[f\left(\xi_p\right)\right]^2}\right)$$

Proof. Since

$$\begin{aligned}
P\left(\sqrt{n}(\hat{\xi}_{pn} - \xi_p) \le t\right) &= P\left(p \le F_n(\xi_p + \frac{t}{\sqrt{n}})\right) \\
&= P\left(\sum_{i=1}^{n} I\left(X_i > \xi_p + \frac{t}{\sqrt{n}}\right) \le n(1-p)\right) \\
&= P\left(\frac{\sum_{i=1}^{n} V_{in} - n\bar{F}(\xi_p + \frac{t}{\sqrt{n}})}{\sqrt{n\bar{F}(\xi_p + \frac{t}{\sqrt{n}})[1 - \bar{F}(\xi_p + \frac{t}{\sqrt{n}})]}} \le \frac{n(1-p) - n\bar{F}}{\sqrt{n\bar{F}(1-\bar{F})}}\right) \\
&\to \Phi\left(\frac{tf(\xi_p)}{\sqrt{p(1-p)}}\right)
\end{aligned}$$

□